CNN BASIC

DISCLAIMER

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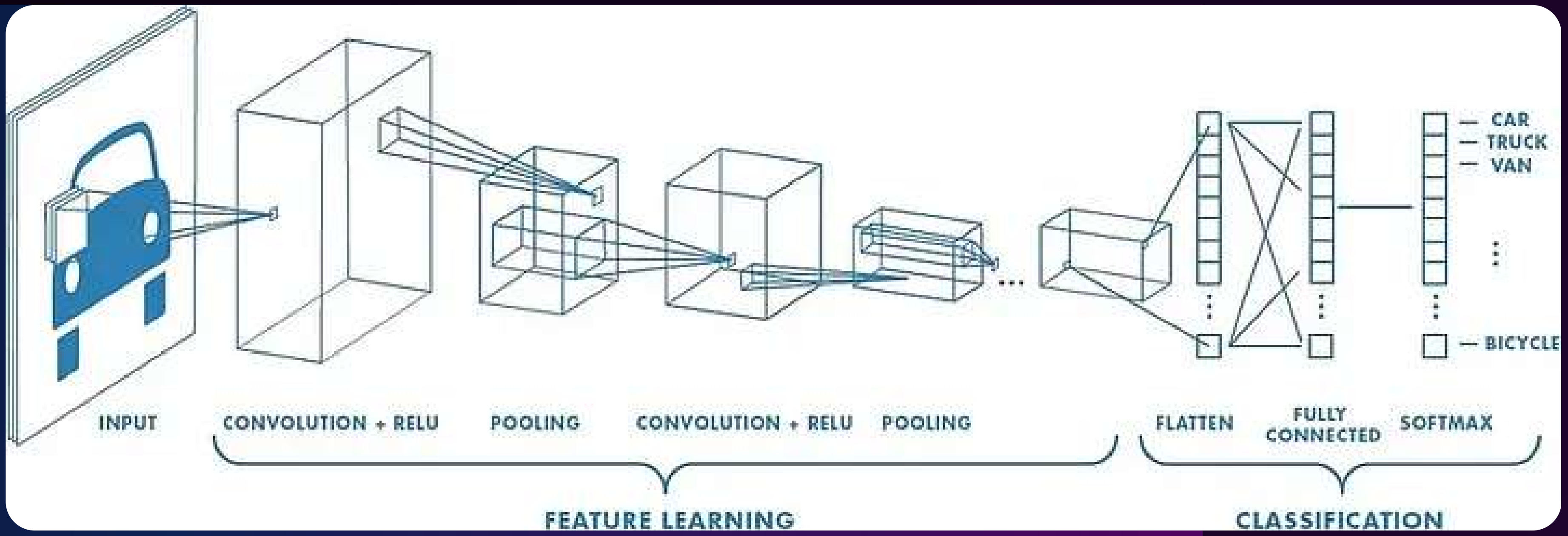
Convolutional Neural Networks



What is CNN?

- A convolutional neural network or CNN is a type of artificial neural network used in image recognition and processing that is inspired by the biological processes in the visual cortex of animals. They are made up of neurons that have learnable weights and biases.
- CNNs use a technique called convolution instead of general matrix multiplication in at least one of their layers. Convolution is a specialized kind of linear operation.

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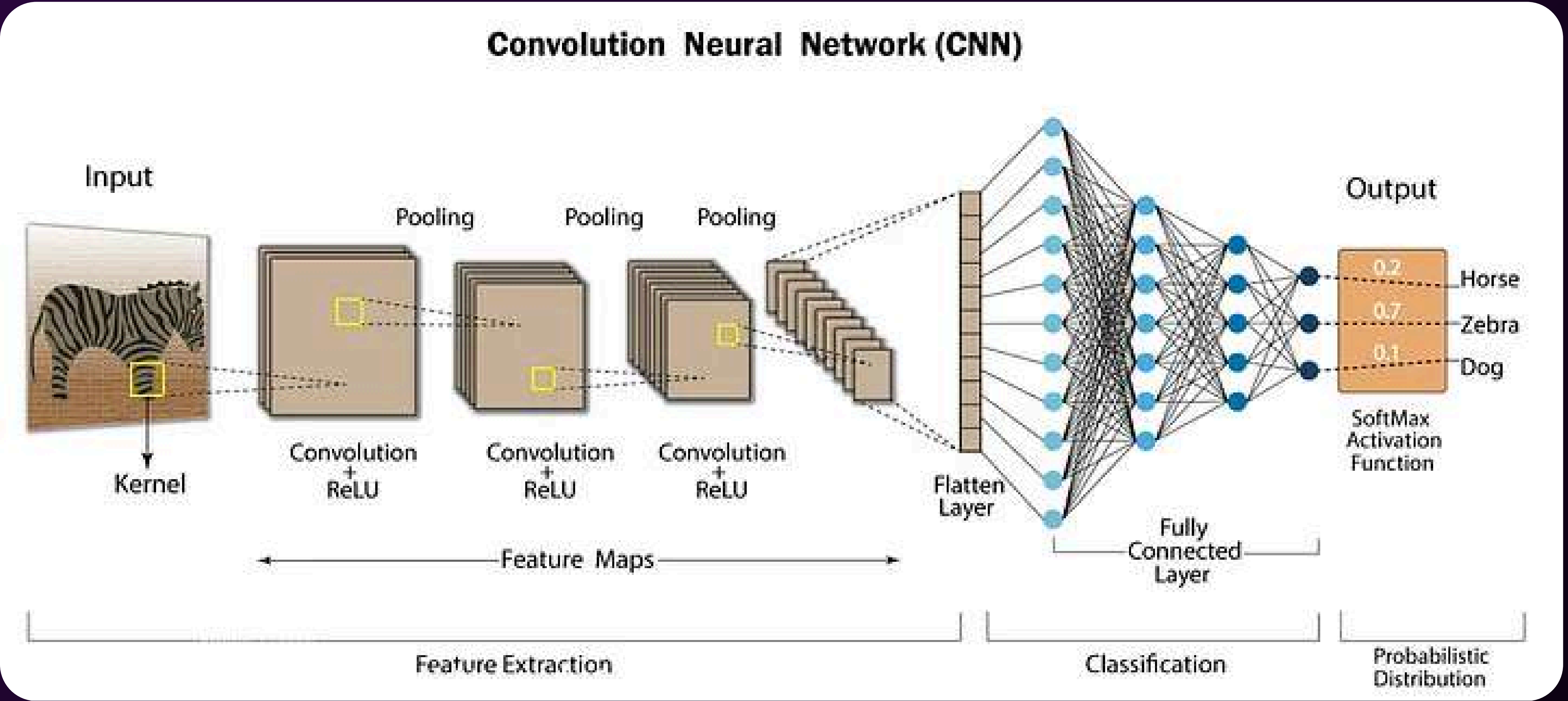
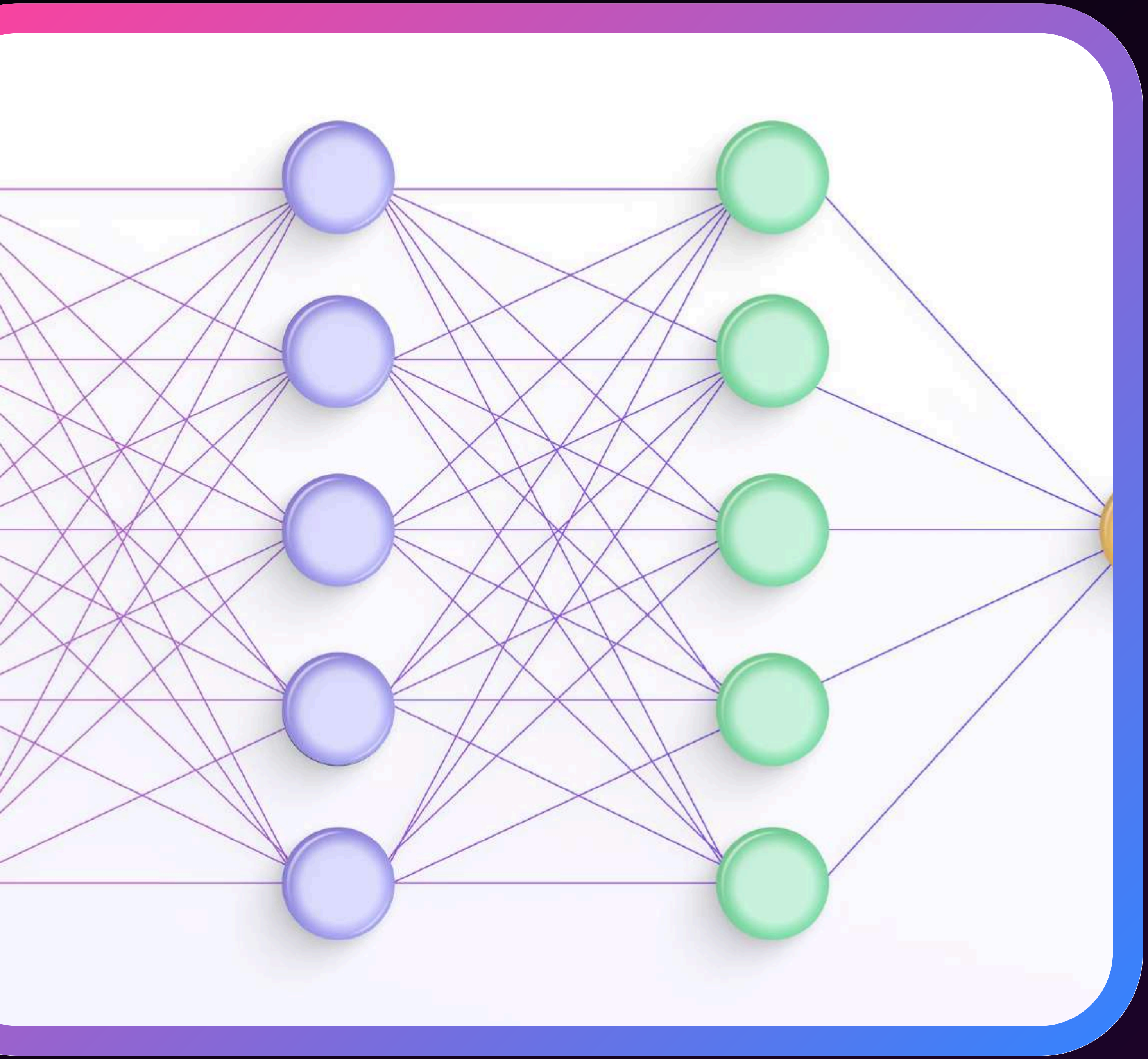
Convolutional Neural Network

How it works



Convolutional Neural Networks (CNNs) process images by using small filters to detect features, creating activation maps. These maps are then downsampled by pooling layers for efficiency. Finally, a fully connected layer classifies the image. Popular CNN architectures include AlexNet, VGGNet, ResNet, and Inception, which excel in tasks like object identification and medical image analysis. Building a CNN involves choosing hyperparameters and training the network on large datasets using backpropagation. With sufficient resources, CNNs can surpass human capabilities in various visual tasks and are widely used in image recognition applications.

Layers of Convolutional neural network:

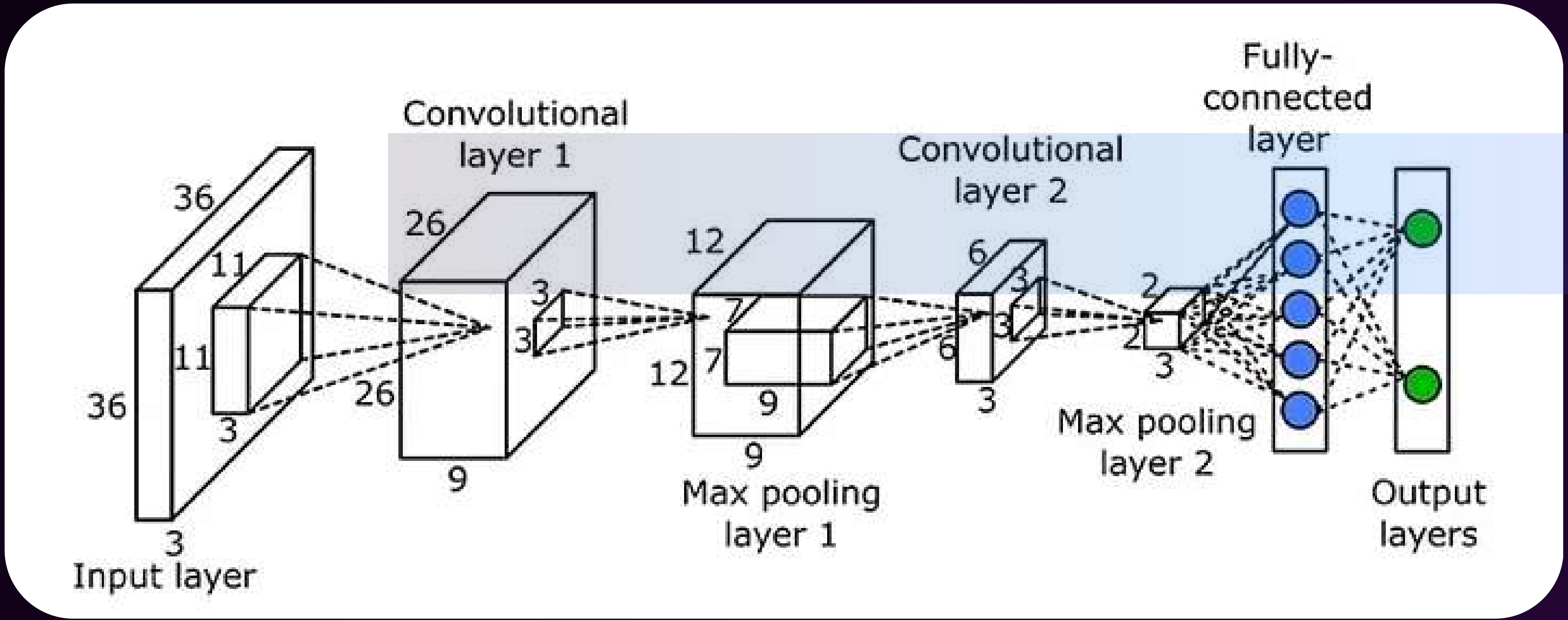


6 Layers

- Convolutional Layers
- Pooling Layers
- Activation Layer
- Normalization Layer
- Dropout Layer
- Dense Layer

Convolutional Layers

WHAT IS THIS?



Convolutional layers in CNNs use filters that slide across an image to perform a convolution operation. This process detects features such as edges and curves, with multiple filters identifying diverse characteristics. The convolution results in a feature map highlighting the location and intensity of detected features. Stacking these layers allows the network to learn increasingly complex and high-level features.

MORE

FILTERS (KERNELS)

CNN filters are small, adjustable matrices learning to extract specific image features, optimized during training.

STRIDE

Stride dictates filter movement in convolution; larger steps yield smaller, faster-computed feature maps.

PADDING

Padding adds pixels to control output size; 'valid' is none, 'same' matches input dimensions.

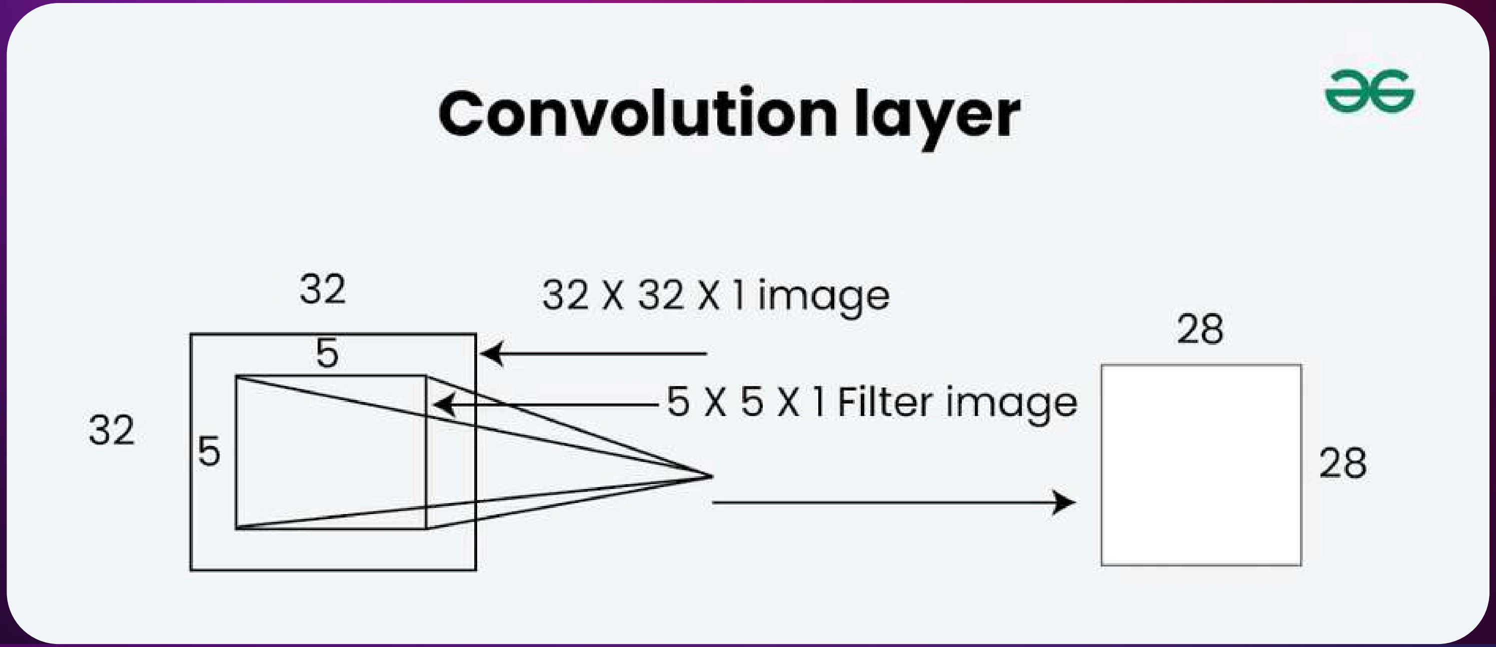
ACTIVATION FUNCTION

After convolution, ReLU activation adds non-linearity, enabling the network to learn complex data patterns.

Convolution Layer

Processing Steps

1. **Initialize Filters**
   - Randomly initialize a set of filters with learnable parameters.
2. **Convolve Filters with Input**
   - Slide the filters across the width and height of the input data, computing the dot product between the filter and the input sub-region.
3. **Apply Activation Function**
   - Apply a non-linear activation function to the convolved output to introduce non-linearity.
4. **Pooling (Optional)**
   - Often followed by a pooling layer (like max pooling) to reduce the spatial dimensions of the feature map and retain the most important information.



Example of Convolution layer

Benefit

PARAMETER SHARING

The same filter is used across different parts of the input, reducing the number of parameters and computational cost.

LOCAL CONNECTIVITY

Each filter focuses on a small local region, capturing local patterns and features.

HIERARCHICAL FEATURE LEARNING

Multiple convolution layers can learn increasingly complex features, from edges and textures in early layers to object parts and whole objects in deeper layers.

Pooling Layers

WHAT IS THIS?

Why important?

DIMENSIONALITY REDUCTION

Pooling layers reduce the spatial size of the feature maps, which decreases the number of parameters and computations in the network. This makes the model faster and more efficient.

TRANSLATION INVARIANCE

Pooling helps the network become invariant to small translations or distortions in the input image. For example, even if an object in an image is slightly shifted, the pooled output will remain relatively unchanged.

OVERFITTING PREVENTION

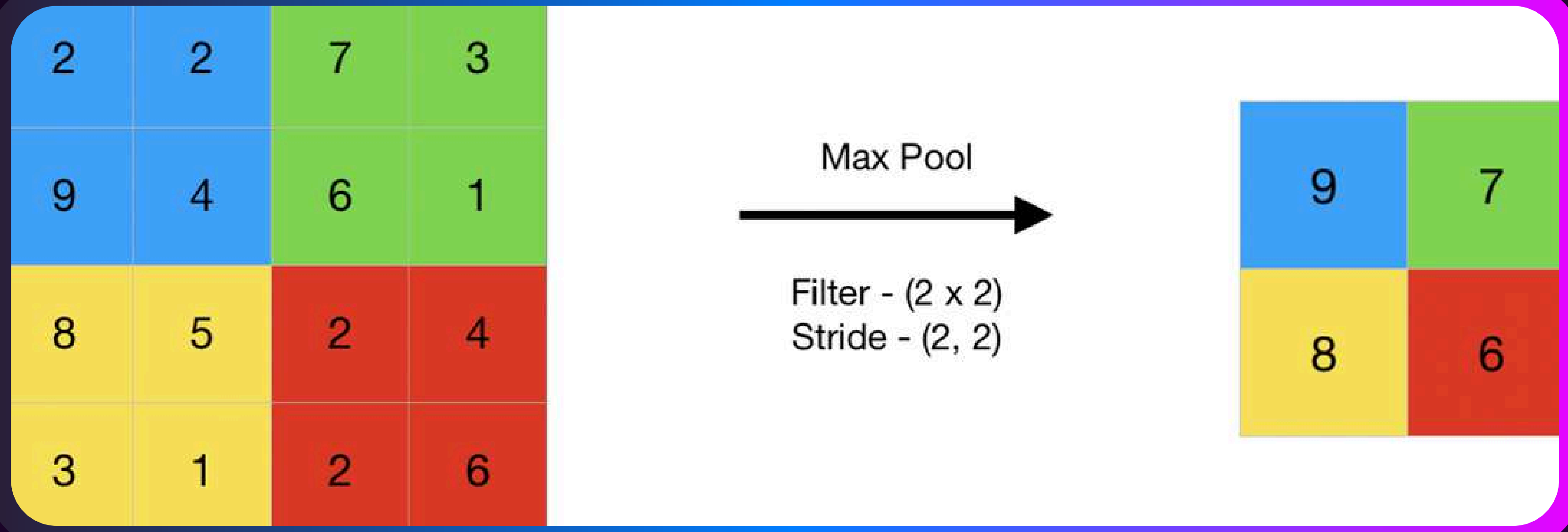
By reducing the spatial dimensions, pooling layers help prevent overfitting by providing a form of regularization.

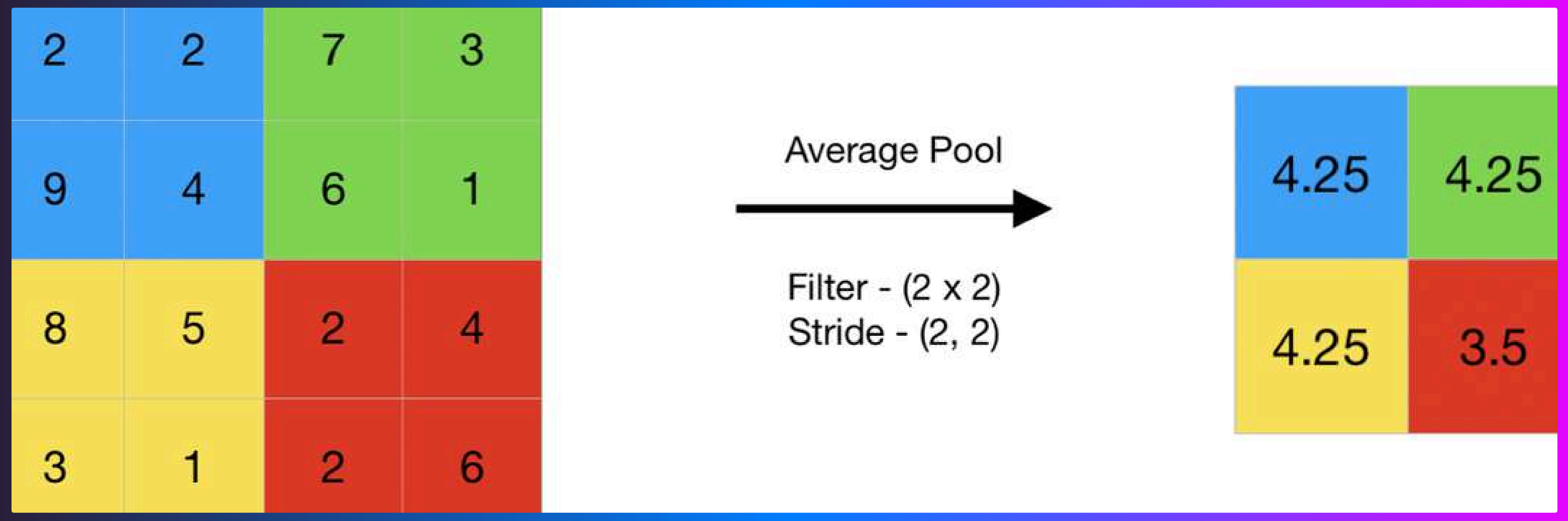
FEATURE HIERARCHY

Pooling layers help build a hierarchical representation of features, where lower layers capture fine details and higher layers capture more abstract and global features.

Max Pooling

Max pooling selects the strongest features, creating a map of the most prominent information, often improving performance.





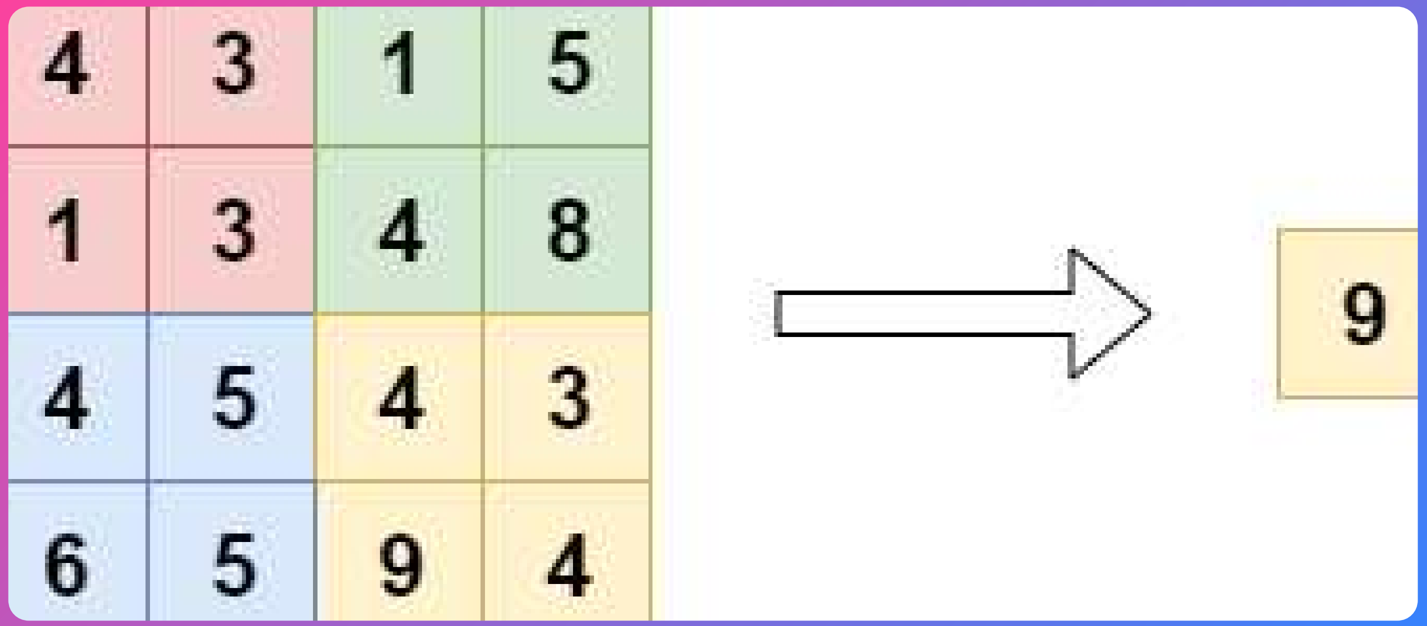
Average Pooling

Average pooling computes the mean of features in a region, offering a generalized representation useful for preserving overall context.

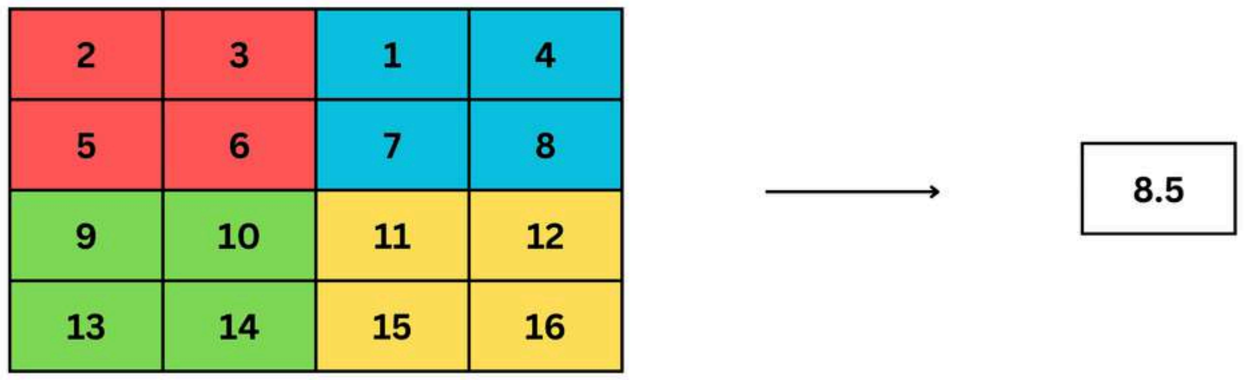
Global Pooling

Global pooling reduces each feature map channel to a single value (1x1xnc) using max or average across the entire map.

Global Max Pooling



Global Average Pooling



No Conecpt of Windows. Take Average of all Values in the Matrix.

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Activation Layers

WHAT IS THIS?

Types

Of Activation Functions

LINEAR ACTIVATION FUNCTION

Linear activation: output is a linear combination of input; limited learning.

NON-LINEAR ACTIVATION FUNCTIONS

SIGMOID FUNCTION

Sigmoid: 'S' shaped, output 0 to 1, good for binary classification.

TANH ACTIVATION FUNCTION

Tanh: shifted sigmoid, output -1 to +1, common in hidden layers.

RELU (RECTIFIED LINEAR UNIT) FUNCTION

ReLU: output is max(0, x), non-linear, efficient for computation.

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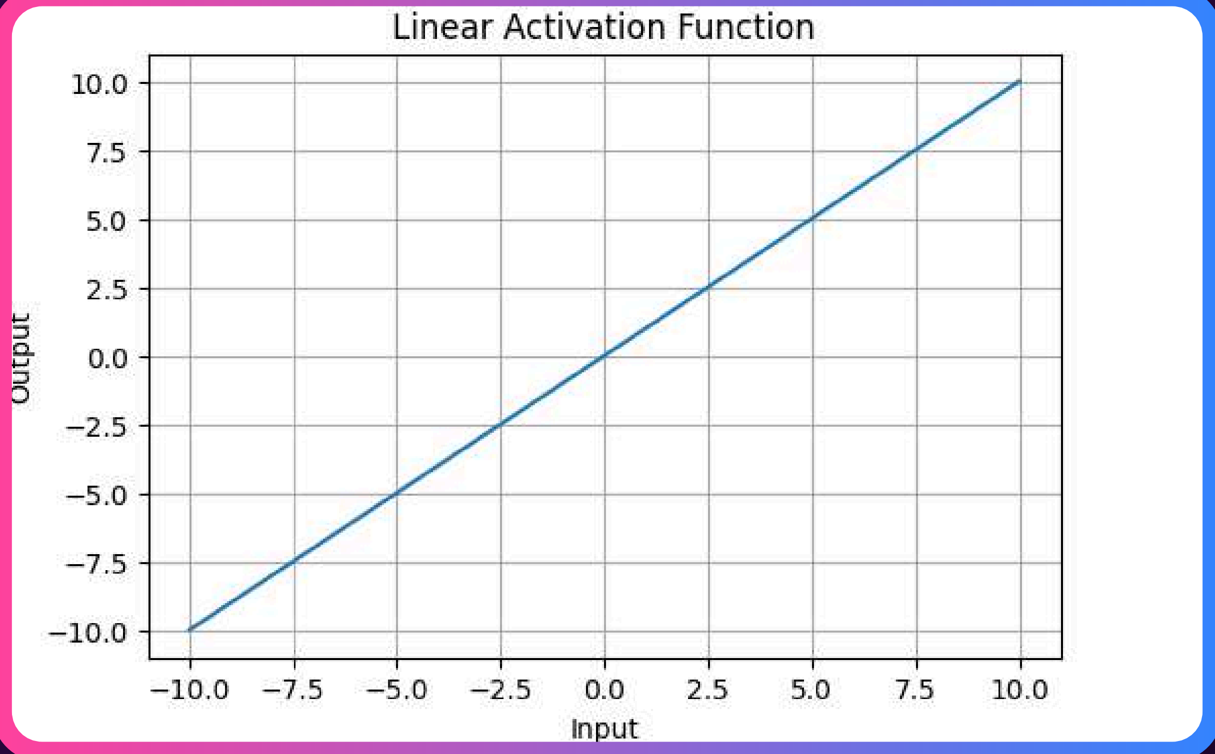

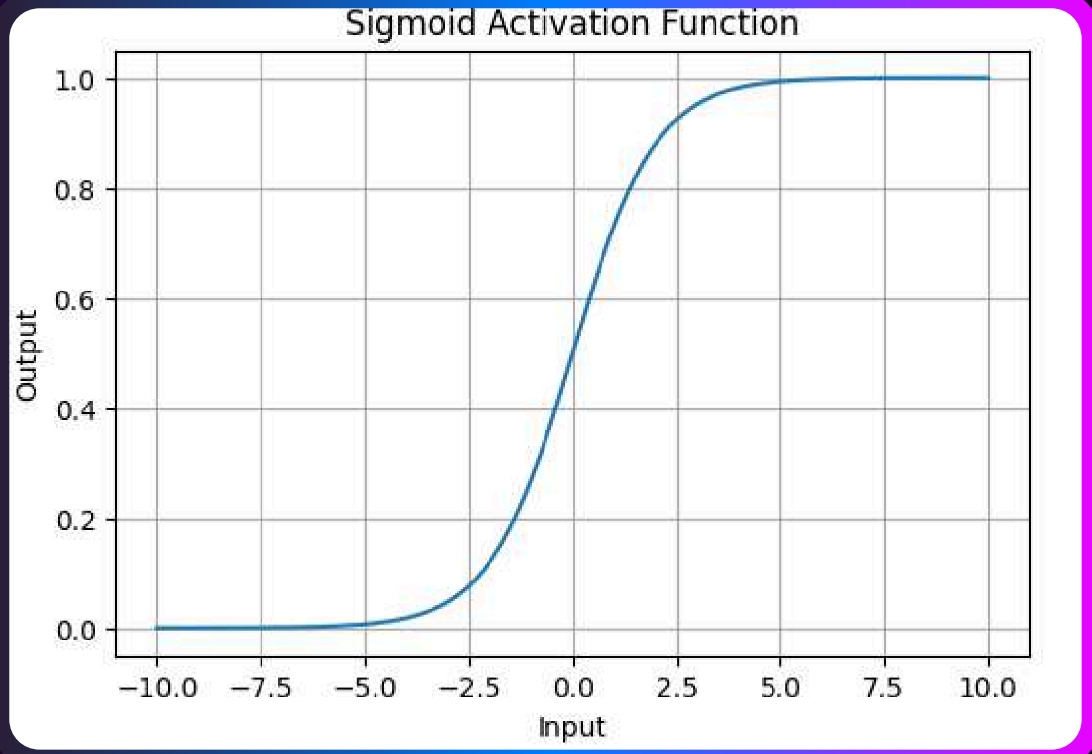

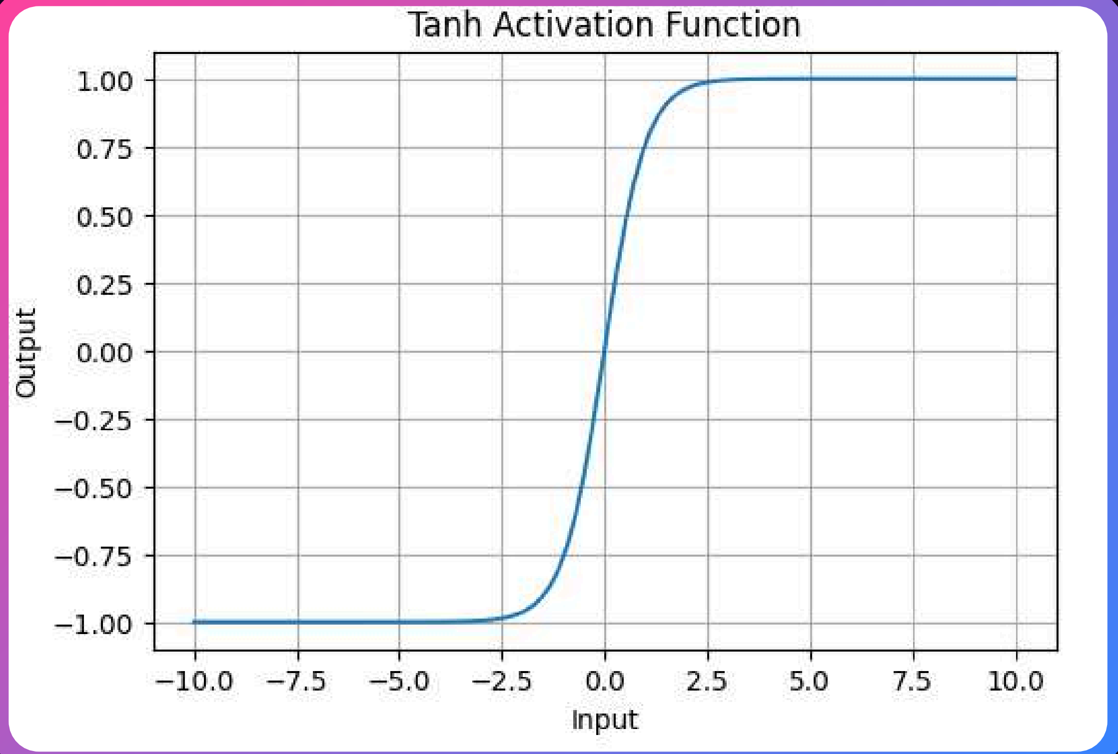

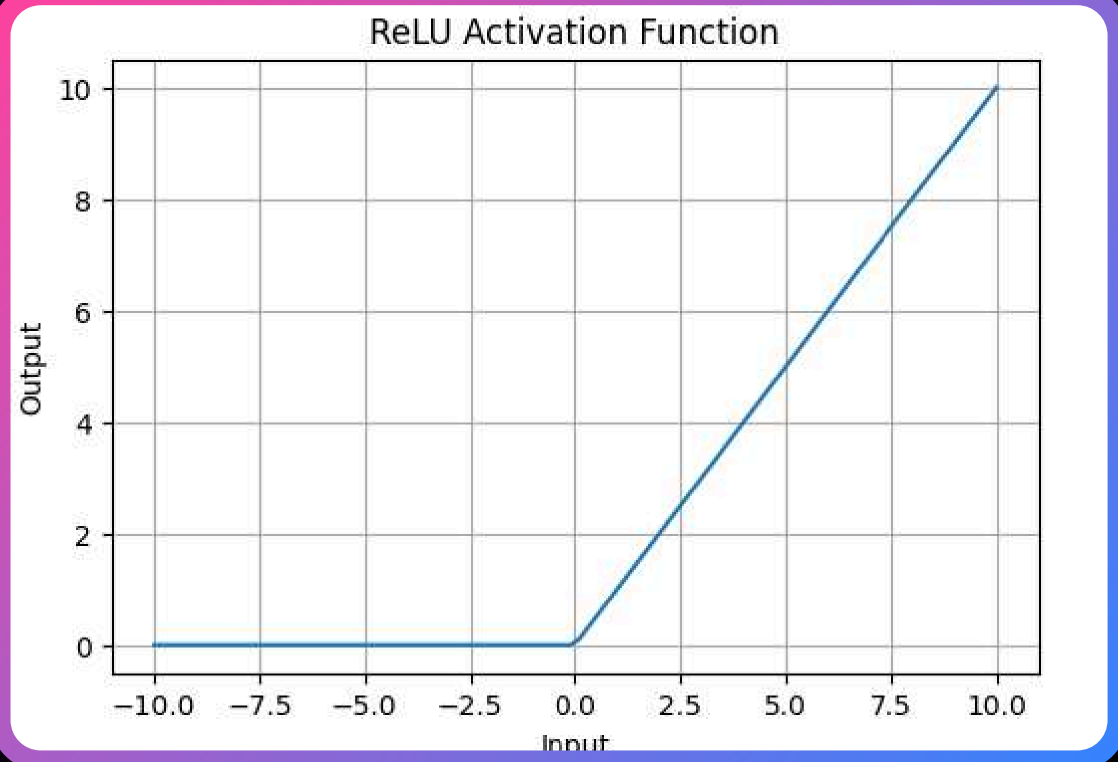

ACTIVATION

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Types

Of Activation Functions

EXPONENTIAL LINEAR UNITS

SOFTMAX FUNCTION

Softmax: converts scores to probabilities (0-1) for multi-class classification.

SOFTPLUS FUNCTION

Softplus: smooth, non-linear, positive output (0 to ∞), differentiable alternative to ReLU.

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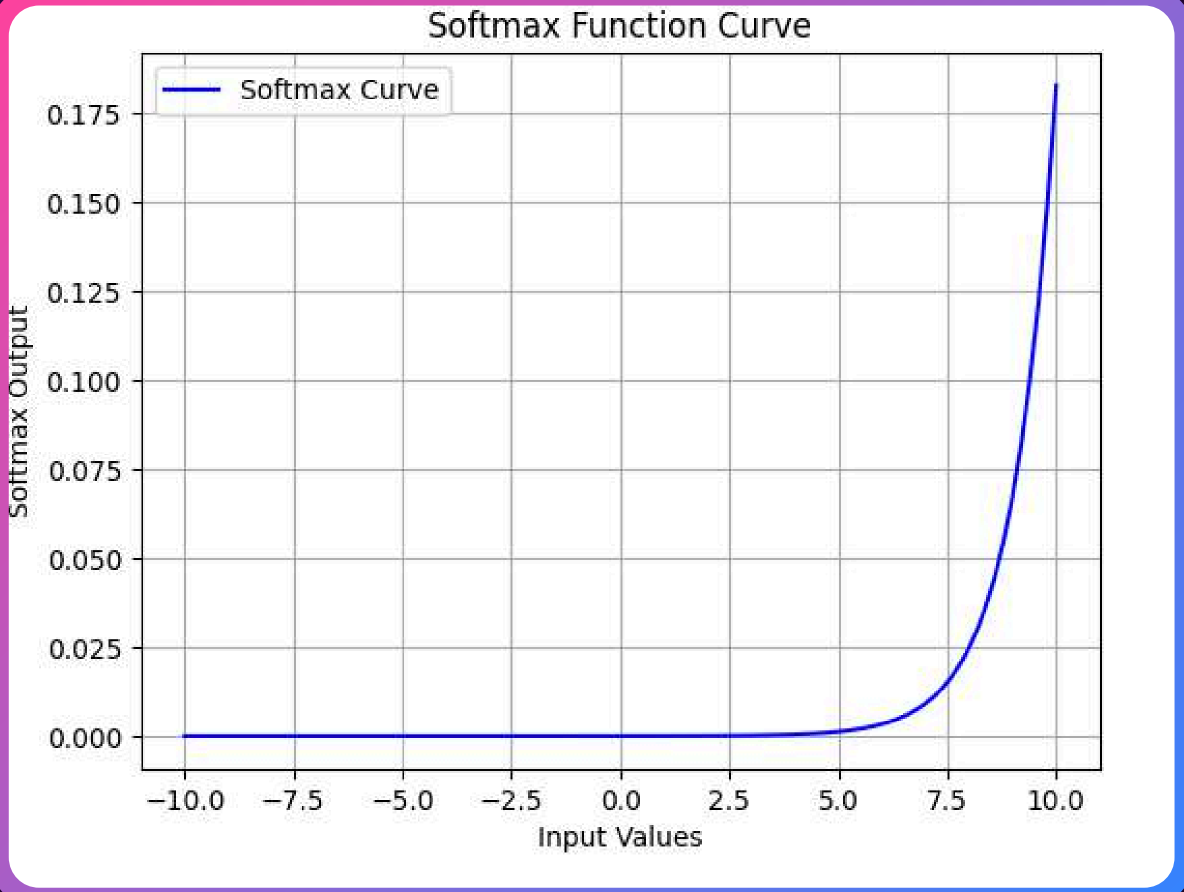
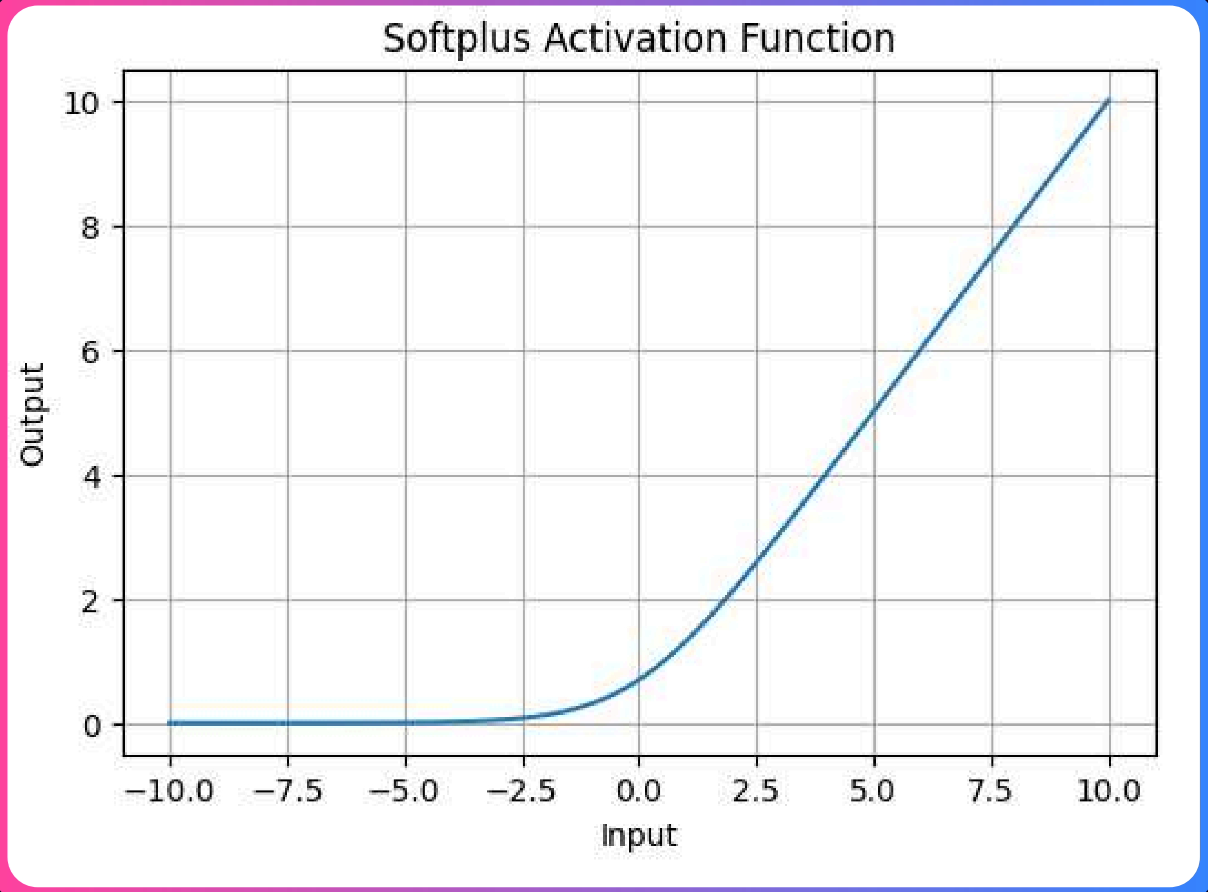




ACTIVATION

Impact of Activation Functions on Model Performance

ONVERGENCE SPEED

Functions like ReLU allow faster training by avoiding the vanishing gradient problem, while Sigmoid and Tanh can slow down convergence in deep networks.

GRADIENT FLOW

Activation functions like ReLU ensure better gradient flow, helping deeper layers learn effectively. In contrast, Sigmoid can lead to small gradients, hindering learning in deep layers.

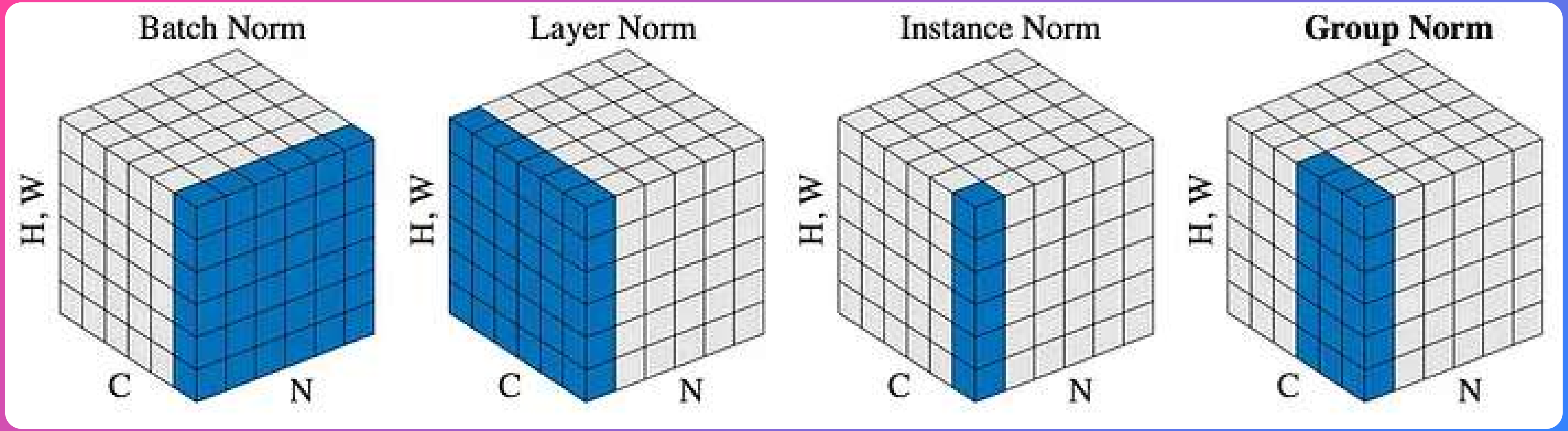
MODEL COMPLEXITY

Activation functions like Softmax allow the model to handle complex multi-class problems, whereas simpler functions like ReLU or Leaky ReLU are used for basic layers.

Normalization Layer

WHAT IS THIS?

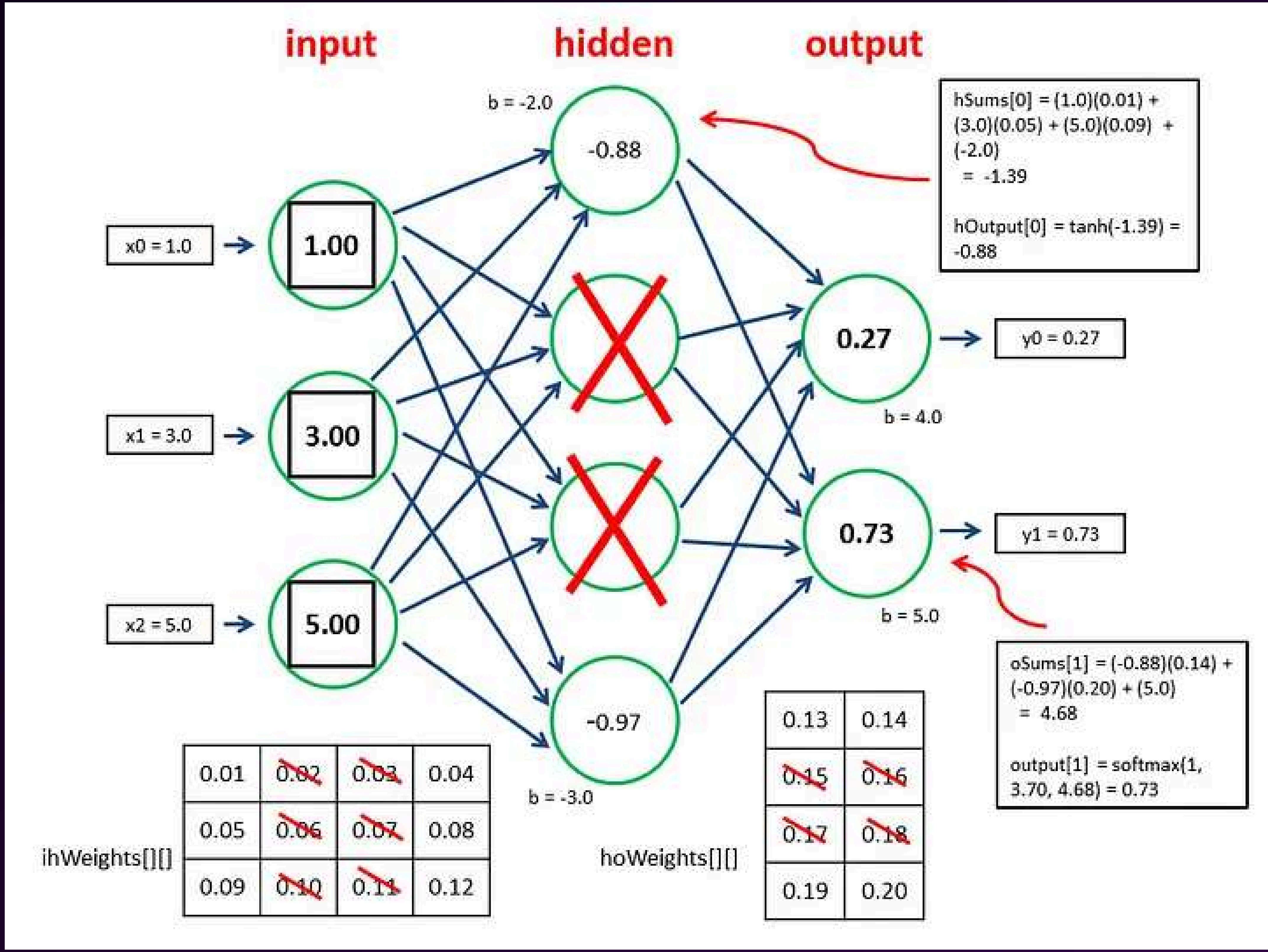
The normalization layer performs normalization operations, such as batch normalization or layer normalization, to ensure that the activations of each layer are well-conditioned and prevent overfitting.



Dropout Layer

WHAT IS THIS?

The dropout layer is used to prevent overfitting by randomly dropping out neurons during training. This helps to ensure that the model does not memorize the training data but instead generalizes to new, unseen data.



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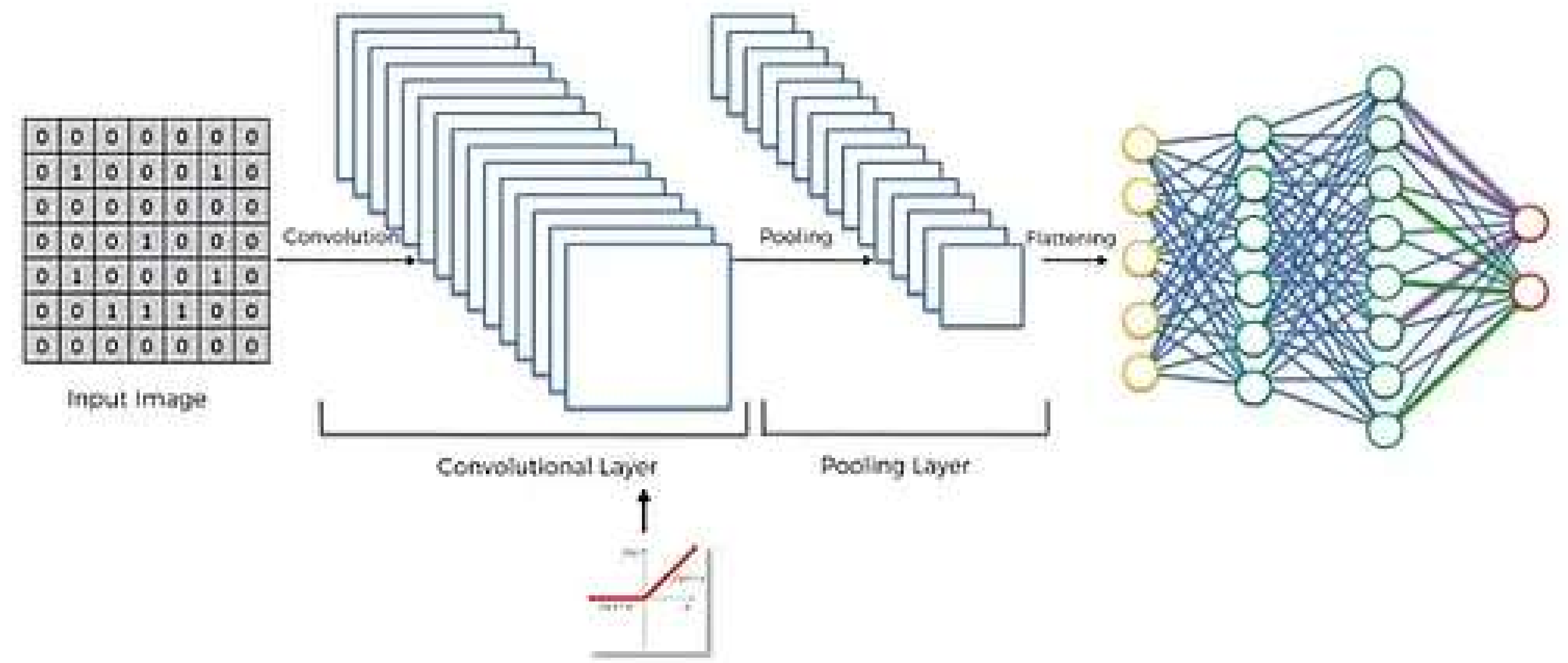
Dense Layer

WHAT IS THIS?

Dense layers in CNNs combine extracted features for the final prediction. Activations are flattened and fed into the dense layer, which uses weighted sums and activation functions for the output.



BENEFITS OF CNN:

FEATURE EXTRACTION

CNNs automatically learn and extract relevant image features, decreasing the need for manual feature design.

SPATIAL INVARIANCE

CNNs excel at object recognition by identifying objects despite variations in location, size, and orientation within an image.

ROBUST TO NOISE

CNNs exhibit robustness by effectively processing noisy or cluttered images, crucial for diverse real-world applications with varying image quality.

PERFORMANCE

CNNs achieve top performance in computer vision tasks like image classification, object detection, and semantic segmentation.

TRANSFER LEARNING

Pre-trained CNN models can be reused, saving data and computation for new tasks through transfer learning.

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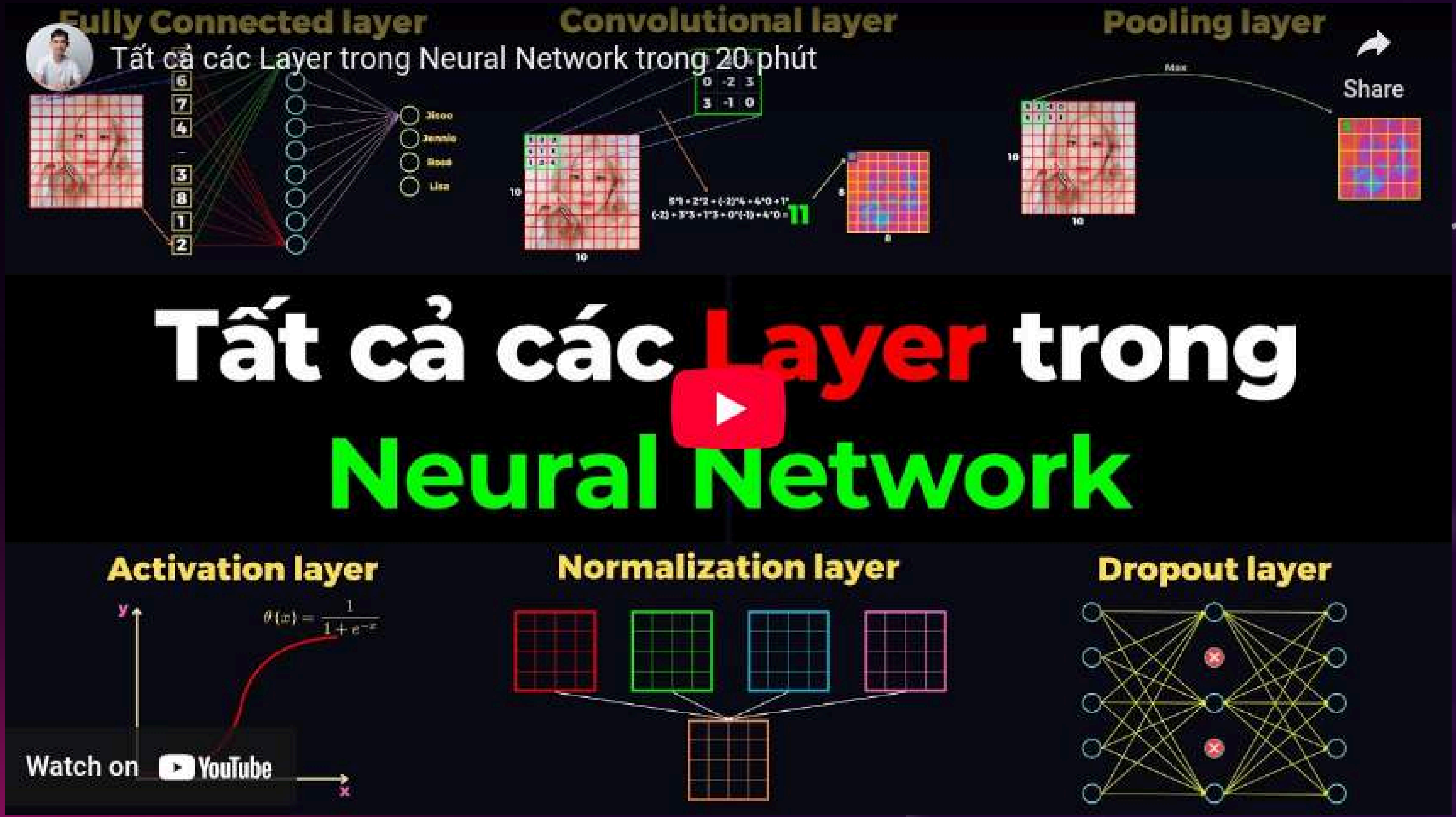

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